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Digital humanities and archival studies: a cross-linguistic analysis

Chyi-Kuan Wang and Chiao-Min Lin
Graduate Institute of Library, Information and Archival Studies, National Chengchi University, Taipei City, Taiwan

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Abstract

Purpose – This study aims to examine how digital humanities methodologies, particularly natural language processing and network analysis, have influenced archival scholarship across Chinese- and English-language contexts from 2004 to 2023.

Design/methodology/approach – The study uses latent Dirichlet allocation for topic modeling, term frequency-inverse document frequency for keyword extraction and social network analysis to examine thematic patterns in archival journal articles. Chinese knowledge information processing tagger is used for Chinese-language data, while Gensim with Chinese and Gensim natural language toolkit is applied to English-language data, enabling cross-linguistic comparison of thematic patterns.

Findings – The results reveal distinct thematic divergence. Chinese-language journals emphasize state-led initiatives, such as national identity construction and digital infrastructure, whereas English-language journals focus on community archives, Indigenous rights and archival justice. Despite these differences, both corpora converge on themes such as digital archive management, policy dissemination and archival education, reflecting a shared emphasis on archival infrastructure and governance. Topic modeling identifies six coherent themes in Chinese articles and seven in English. Keyword analysis shows that Chinese literature prioritizes institutional roles, whereas English texts emphasize social justice and community engagement.

Originality/value – The study's primary contribution lies in its cross-linguistic comparative design, which integrates topic modeling, keyword clustering and social network analysis across Chinese- and English-language scholarly corpora. This approach highlights divergent orientations in archival knowledge production across the two communities, a dimension that has received limited systematic attention in prior single-language or single-method studies.

Keywords Digital humanities, Archival scholarship, Cross-linguistic analysis, Topic modeling, Keyword clustering

Paper type Research paper

Introduction

Since the early 2000s, the digitization of archival collections and the emergence of born-digital records have substantially transformed how archival content is analyzed, preserved and accessed. These developments have prompted archival scholars and practitioners to reconsider foundational assumptions about what records are and how they function. Computational archival science, for instance, conceptualizes records as data amenable to algorithmic processing, marking a significant departure from earlier archival epistemologies (Marciano *et al.*, 2018). More broadly, Berry (2011) characterizes this reorientation as part of a *Computational Turn*, in which digital technologies reshape not only scholarly methods but also the objects and evidentiary standards that constitute knowledge. Archival studies have actively participated in this shift by adopting computational and text-analytical tools to reinterpret historical materials. Digital humanities (DH), drawing on humanities computing and corpus linguistics (McCarty, 2005), has provided much of the methodological infrastructure for these developments. Around 2010, its "second wave" introduced a stronger linguistic orientation, foregrounding textual complexity, media specificity and interpretive context (Hayles, 2012; Parikka, 2012).

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The implications of this linguistic turn for archival practice are substantial. Long *et al.* (2025) identify four interconnected developments: the expansion of archival terminology to accommodate digital materials, the application of archival principles to born-digital content management, the integration of archival perspectives into (DH) projects and the repositioning of archives within broader social and digital contexts. Collectively, these shifts have expanded the role of digital archives in the construction of historical knowledge (Long, 2024; Wang *et al.*, 2020). Long (2024) further argues that the linguistic turn enables more systematic modes of textual analysis, making it possible to trace how archives encode social issues and shape representations of individual identity.

Despite this methodological expansion, the broader conceptual implications of datafication, particularly the interpretive risks associated with treating archival content as computational data, remain underexamined (Southerton, 2022). The increasing use of frequency and co-occurrence analysis in conceptual history and historical semantics illustrates how these methods are being applied in practice (Long *et al.*, 2021; Long *et al.*, 2025; Schwandt and Wachter, 2024). A more fundamental gap, however, concerns the cross-linguistic dimension of this transition. Cook (2013) documents a community-centered shift in English-language archival theory, yet systematic comparisons of how computational methods are adopted and framed across different linguistic scholarly communities remain limited. English-language archival literature tends to foreground technical implementation, with computer science–related topics accounting for over 28% of the field's output (Hawkins, 2022; Lu and Ling, 2024). In contrast, Chinese-language scholarship has examined thematic evolution through keyword frequency analysis, institutional collaboration mapping and citation network analysis (Guo and Di, 2019; Lin, 2013; Yan *et al.*, 2026). These differing orientations warrant cross-contextual examination.

This study addresses this gap by applying an integrated computational framework that combines latent Dirichlet allocation (LDA) topic modeling, term frequency-inverse document frequency (TF-IDF) weighting and keyword co-occurrence network analysis to a corpus of archival scholarship published in Chinese and English between 2004 and 2023. The analysis traces the adoption of computational approaches across geopolitical and linguistic contexts and maps the evolution of associated knowledge structures over time.

Research purpose and objectives

This study pursues two main objectives:

(1) To map thematic divergence: to compare thematic developments in Chinese- and English-language archival scholarship and identify how regional and linguistic contexts shape the articulation of DH-related topics across a corpus of 37,914 articles published between 2004 and 2023.

(2) To trace methodological evolution: to examine how computational approaches have been adopted within archival research practices over time and to identify temporal patterns in the development of key archival concepts across both scholarly communities.

By comparing thematic patterns and methodological tendencies across these communities, the study identifies variations in cross-disciplinary engagement and traces how archival frameworks have evolved in different linguistic and regional contexts.

Literature review

DH brings together interpretive traditions in the humanities and social sciences with computational tools and digital platforms. Rather than constituting a single unified discipline,

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DH represents an assemblage of diverse practices and methodological orientations (Svensson, 2012). This convergence has enabled large-scale textual analysis, facilitated interdisciplinary collaboration and broadened public engagement with scholarly materials (Brügger, 2016; Hsiang and Tu, 2011). Related fields, particularly library and information science (LIS), have likewise incorporated digital techniques such as keyword analysis and subject classification to examine thematic structures and patterns of knowledge production (Chen *et al.*, 2015). Collectively, these developments provide a methodological basis for extending computational approaches into archival research.

In archival research, digitization has shaped how archives are conceptualized, accessed and interpreted. The availability of digital formats has made techniques such as distant reading more practicable, enabling comparative analysis and visualization of archival materials at larger scales, thus expanding the methodological repertoire available in the humanities. These changes are reflected in shifting themes within archival journals, illustrating broader methodological transitions across two main phases in the development of DH (Cox, 1987; Craig, 1996; Gilliland and McKemmish, 2004, 2018; McKemmish *et al.*, 2016). The first phase introduces computational and quantitative methods, including database querying, corpus linguistics and hypertext frameworks. The second phase emphasizes qualitative and interpretive approaches, such as experiential, interactive and generative methodologies that integrate digital tools with core humanities research objectives (Zhang *et al.*, 2024).

Since 1949, scholars have explored the thematic development of archival research in Chinese literature (Guo and Di, 2019; Lin and Fan, 2010; Zhang and Yang, 2020). More recently, Feng *et al.* (2021) analyzed articles indexed in the Chinese social sciences citation index (CSSCI) during China's 13th Five-Year Plan period (2016–2020). They identified six major themes: archival theory, digital transformation, system reform, preservation and service, diversified management models and the historical development of archival studies in China. These themes provide a structured lens for understanding disciplinary trends in the Chinese context. In contrast, English-language studies have emphasized interdisciplinary collaboration and the evolving professional roles of archivists (Cook, 2013; Gilliland and McKemmish, 2018). Reviewing major research trends over the past three decades, Gilliland and McKemmish (2018) note that technological and educational changes, combined with broader social and intellectual movements, have prompted archivists to engage more closely with other academic disciplines, government institutions, corporations, nonprofits and local communities. These observations highlight the growing recognition of cross-sector collaboration in archival studies.

Technological advancements have contributed to a thematic shift in archival research, moving beyond traditional historical concerns to include topics such as social justice, human rights and the documentation of marginalized communities (McKemmish *et al.*, 2016). The rise of social media and digital communication tools has further diversified research agendas, contributing to the notion of an "archival multiverse." Alongside these changes, archivists are increasingly regarded as both authoritative gatekeepers and facilitators of shared memory, promoting the inclusion of multiple perspectives across time, space and cultural contexts (Cook, 2013).

Previous studies have used various computational techniques in archival research, including corpus-driven analysis and network-based models (Couture and Ducharme, 2005; Cox, 1987; Craig, 1996; Gilliland and McKemmish, 2018; Guo and Di, 2019; Hsueh and Wang, 2017; Lin, 2013; Lin, 2021; Lin and Fan, 2010; McKemmish *et al.*, 2016; Yan *et al.*, 2012; Zhang and Yang, 2020). These efforts demonstrate the potential of digital tools to analyze scholarly networks and better understand the dynamics of archival knowledge production. Accordingly, analyzing archival journals provides a useful entry point for assessing current trends in DH and textual analysis. Network-based tools have been used to

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visualize academic relationships among authors, journals and disciplines, offering insights into the collaborative structures of archival research.

While existing bibliometric and computational studies of archival scholarship have advanced understanding of publication trends and thematic patterns, they have largely focused on single-language corpora, isolated case studies or narrow temporal windows. Song *et al.* (2023), for instance, acknowledge that their keyword analysis across 18 journals constrained the depth and representativeness of their findings. Wu *et al.* (2024), while attempting cross-linguistic retrieval, encountered what Zhu (2025) describes as "discourse system" gaps between Chinese and international scholarship – structural divergences that keyword-based approaches alone are not designed to bridge. Cross-linguistic comparisons that systematically examine how archival research frameworks develop across distinct linguistic and institutional environments therefore remain rare.

The present study addresses this gap by applying a computational framework that integrates LDA topic modeling, TF-IDF weighting and keyword co-occurrence network analysis to a corpus of 37,914 Chinese- and English-language articles published between 2004 and 2023. In doing so, it aims to map thematic trajectories and identify divergent patterns of knowledge production across the two scholarly communities.

Methodology

This study examines thematic trends in journal articles published between January 2004 and December 2023, drawing on a corpus of Chinese- and English-language archival scholarship.

Standardization procedures were implemented to address differences in metadata structures between the Chinese- and English-language databases. These procedures included the unification of field formats (e.g. titles and author keywords) and the application of language-specific preprocessing pipelines to ensure thematic consistency.

The study period begins in 2004 for two reasons. First, the volume of digitally published journals expanded substantially after 2000, resulting in broader database coverage of electronic publications from 2004 onward. Second, the 2004 edition of *A Guide to the Core Journals of China* placed greater emphasis on digital resources – such as library digitization, online information platforms and born-digital content management – compared to its 2000 edition (Wang, 2014), reflecting the acceleration of China's academic digitization initiatives.

Chinese-language data were collected from the NCL Taiwan Periodical Literature and CNKI Overseas databases. Journal selection followed two official evaluation frameworks: Taiwan's *Humanities and Social Sciences Core Journals List*, evaluated by the National Science and Technology Council and the 10th edition of *A Guide to the Core Journals of China*, compiled by Peking University Library. The latter was preferred over the CSSCI because it includes a dedicated category for archival studies, providing broader coverage of how Chinese archival scholars have engaged with DH across research, teaching and professional writing contexts. After removing duplicate entries and articles lacking abstracts or keywords, 36,750 Chinese-language articles were retained.

English-language data were retrieved from the Web of Science database across three citation indices:

(1) the social sciences citation index;

(2) arts and humanities citation index; and

(3) the emerging sources citation index.

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These indices were supplemented with bibliographic metadata from Scopus, yielding a total of 1,164 articles. The journals included in each subcorpus are listed in Table 1, along with their respective languages, sample sizes and analytical tools applied. In total, 37,914 journal articles were analyzed across the study period (2004–2023). The full screening process is summarized in Figure 1.

The two subcorpora represent structurally distinct publication tiers. They are therefore not treated as directly comparable representations of a single global field, but rather as distinct publication systems shaped by different institutional and indexing conditions. The English-language subcorpus captures the internationally indexed segment of archival publishing, comprising journals with established citation profiles within the Web of Science framework. The Chinese-language subcorpus represents a broader cross-section, spanning core nationally indexed journals as well as regionally distributed practitioner outlets.

The 31:1 volume asymmetry (36,750 versus 1,164 articles) reflects at least two structural features of the respective publishing systems. First, there is a difference in publication tempo: Chinese-language journals typically publish on a monthly or bimonthly basis, whereas English-language journals tend to publish quarterly or biannually. This discrepancy partially accounts for the observed volume imbalance. Second, the two systems operate under substantially different indexing infrastructures and institutional investment scales (Wang, 2014). This asymmetry is therefore treated as a characteristic of the publishing systems rather than a methodological limitation. Accordingly, the thematic findings should be interpreted as characterizing these specific publication tiers rather than the entirety of archival scholarship in either language.

LDA was used to identify latent semantic structures within the Chinese- and English-language corpora. Topic modeling was implemented using the Machine Learning for Language Toolkit (MALLET) via the Gensim wrapper, which estimates model parameters through Gibbs sampling. This implementation was selected over standard variational Bayes approaches, as Gibbs sampling has been shown to produce more coherent and stable topic solutions for corpora characterized by high-dimensional vocabularies and linguistic variation (Griffiths and Steyvers, 2004). LDA operates by assessing keyword distributions across a predefined number of topics, using patterns of word co-occurrence to generate interpretable topic clusters (Blei and Lafferty, 2007; Yen and Fu, 2022). To determine the optimal number

Table 1. Selected archival core journals and analytical tools used

Language	Journals	No. of articles	Topic modeling tools	Keyword analysis tools
Chinese	Archives Science Bulletin	2,917	LDA	CKIP Tagger
	Archives Management	5,468		
	Shanxi Archives	4,736		
	China Archives	8,650		
	Beijing Archives	6,735		
	Zhejiang Archives	7,315		
	Archives Semiannual	617		
	Journal of InfoLib and Archives	312		
English	Archival Science	429	Gensim	NLTK
	Archives and Records	280		
	Records Management Journal	455		

Source(s): Authors' own work

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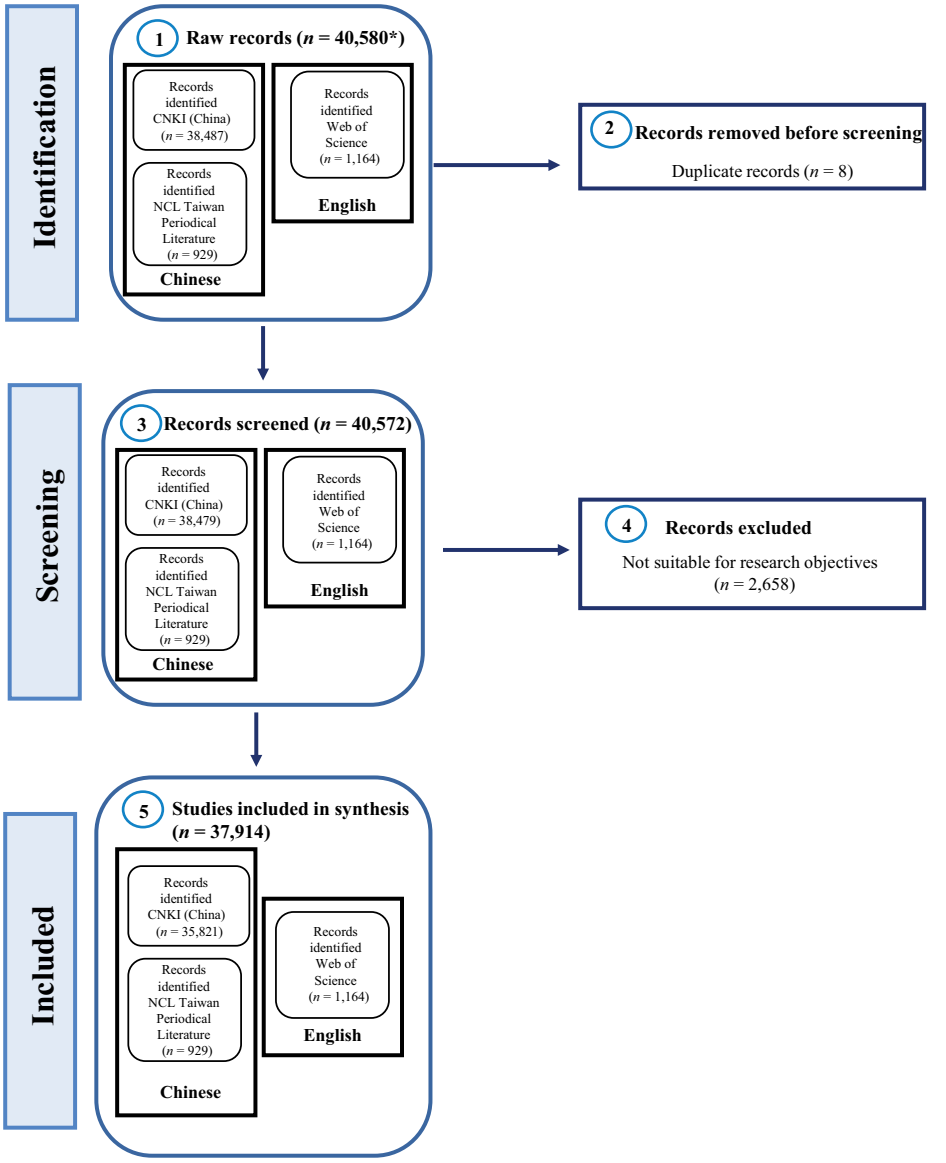


Figure 1. PRISMA flow diagram
Note(s): Filtered by document type: "article"/"research paper"
Source: Authors' own work

of topics for each subcorpus, the model was evaluated using C_V coherence scoring applied to article titles, abstracts and author keywords across the full 2004–2023 period.

Because the two subcorpora were modeled independently, cross-linguistic alignment was conducted through a structured qualitative comparison procedure rather than through

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projection into a shared multilingual embedding space. This methodological choice was deliberate, as projecting both corpora into a shared embedding space risks flattening the structural divergence this study aims to document. For each topic in one subcorpus, the top 20 keywords were compared with the keyword sets of all topics in the other subcorpus. Two topics were classified as aligned if at least six of the top 20 keywords had direct translation equivalents or discipline-recognized functional equivalents (e.g. "數位化/数字化" ≈ "digitization") and if the conceptual domain represented by the topic keyword set was judged to correspond to the same archival practice area. Alignment decisions were reached through structured discussion between the first author and a coauthor with specialist expertise in archival studies, with disagreements resolved through iterative review of the relevant keyword sets.

Following alignment, researcher-led interpretation was used to label and categorize topic clusters based on their most salient keywords. This interpretive process was grounded in a human-in-the-loop topic modeling (HL-TM) framework, which addresses known limitations of fully automated topic models. These limitations include the generation of low-quality topics and the production of topic structures that diverge from domain experts' understanding of a corpus (Hu *et al.*, 2011; Mimno *et al.*, 2011). HL-TM enables domain-informed users to refine model outputs by adjusting word–topic assignments and merging or splitting topics, thereby aligning computational results with established theoretical perspectives in the field (Smith *et al.*, 2018). Following the framework proposed by Hu *et al.* (2011), constraints were interactively applied to encourage selected word sets to co-occur, ensuring that topic labels reflected meaningful conceptual patterns grounded in archival studies.

Language-specific preprocessing tools were used: Chinese knowledge information processing (CKIP) for Chinese and Gensim natural language toolkit (NLTK) for English. In the Chinese corpus, the CKIP tagger was used for segmentation and part-of-speech (POS) tagging due to its superior accuracy in named entity recognition and POS tagging compared to alternatives such as Jieba. For the English corpus, preprocessing was conducted using the Python-based Gensim and NLTK libraries for tokenization, stemming, stop-word filtering and POS tagging.

During preprocessing, Gensim occasionally identified functional words (e.g. "and," "the," "for," "from") as topic-defining terms. To address this issue, NLTK filters, word frequency normalization and synonym clustering techniques were applied to refine thematic specificity. Although different tools were used for the two corpora, POS tagging and keyword normalization procedures were aligned to support meaningful cross-linguistic comparison. The data selection and screening process is summarized in Figure 1 using a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)-style flow diagram.

To address potential analytical bias arising from the volume asymmetry between the two subcorpora, TF-IDF weights were used to identify discriminative terms, with values normalized to mitigate the effects of the 31:1 corpus-size difference. Co-occurrence was defined at the document level. The top 50 terms by TF-IDF weight were selected for co-occurrence network construction, following thresholds established in prior coword studies in information science (Chen *et al.*, 2023; Deng *et al.*, 2021; Ding *et al.*, 2001), which have shown this range to capture core vocabulary without introducing excessive noise from low-frequency terms.

Results

Building on the thematic typologies proposed by Feng *et al.* (2021), Zhang and Yang (2020) and Lin (2021), this study applied topic modeling using CKIP for Chinese texts and NLTK

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for English texts. The results indicate that the optimal number of topics is six for the Chinese corpus (C_V = 0.4346) and seven for the English corpus (C_V = 0.3668), both suggesting relatively high semantic stability. Figures 2 and 3 illustrate the coherence values for the Chinese- and English-language corpora across topic models ranging from 5 to 14 topics.

To enable a broader comparative framework, additional model testing was conducted with topics ranging from 5 to 14 (C_V = 0.4121). This approach facilitated a clearer one-to-one thematic comparison between the two languages. The random state was fixed at 42 to ensure reproducibility and multiple runs of the model produced consistent coherence values across iterations.

A thematic comparison of Chinese and English archival literature reveals three major areas of divergence. Chinese-language journals often explore cross-strait history and cultural identity, particularly in relation to Taiwan–China historical relations and academic collaboration. In contrast, English-language journals emphasize themes such as publishing and theory in the information society, indigenous rights and colonial critique. These themes focus on community memory, indigenous archival claims and the decentralization of archival interpretation.

Thematic comparison of Chinese and English topics

Table 2 presents a thematic comparison of Chinese-and English-language archival literature, revealing three major areas of divergence and five areas of convergence.

Within the Chinese-language corpus, the most prominent divergent theme is "cross-strait history and cultural identity." Archival institutions are depicted as playing a central role in constructing national identity and documenting social transformation. Since the 1990s, the

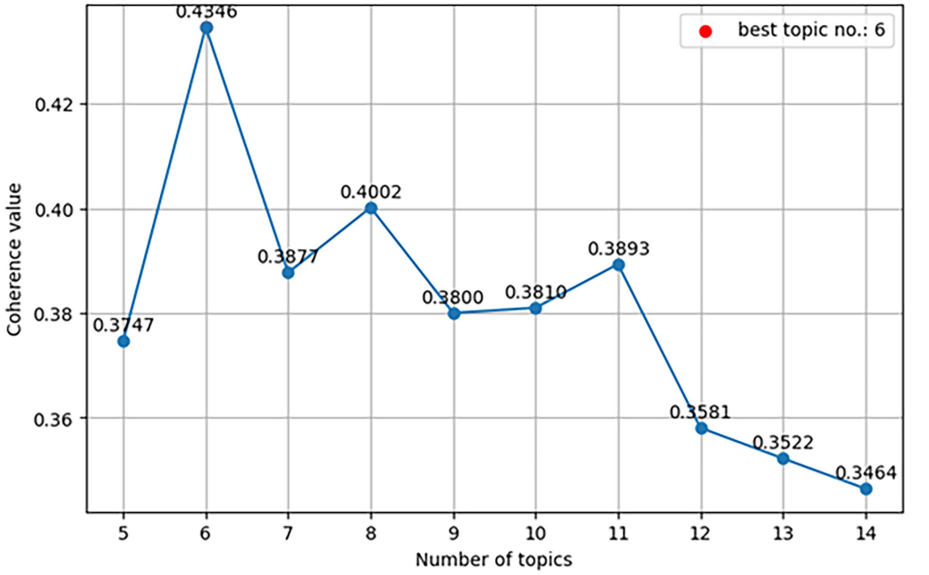


Figure 2. Coherence values (C_V) for the Chinese-language corpus across different numbers of topics (K = 5–14) The optimal solution was identified at K = 6 (C_V = 0.4346)

Source: Authors' own work

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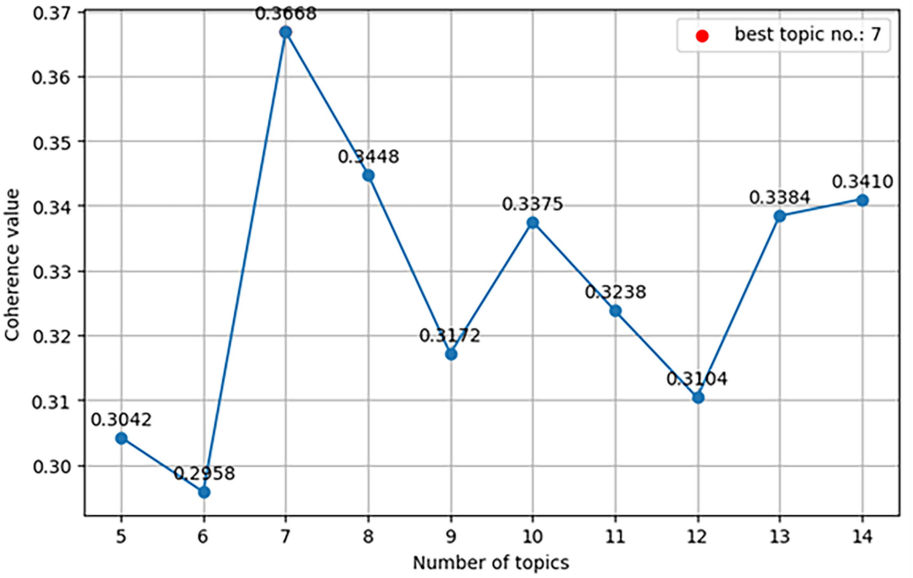


Figure 3. Coherence values (C_V) for the English-language corpus across different numbers of topics (K = 5–14) The optimal solution was identified at K = 7 (C_V = 0.3668)

Source: Authors' own work

growth of digital archives has expanded archival education, while the development of open-access digital platforms has strengthened the role of archives in shaping national narratives (Chang and Yen, 2013; He *et al.*, 2015; Hsu and Fan, 2012). Chinese-language archival journals revisit Taiwan–China relations through the lens of cultural policy and collective memory reconstruction, with archival materials serving as evidence of cross-strait historical linkages.

In contrast, English-language archival literature foregrounds community-based and postcolonial orientations. A prominent theme is "community archives and preservation," in which scholars collaborate with academic libraries and local communities to develop digital heritage projects that prioritize preservation and accessibility (Lucky and Harkema, 2018). A related theme, "Indigenous rights and colonial critique," addresses community memory, Indigenous archival authority and the decentralization of archival interpretation. Community-based archives are framed as serving local needs in ways distinct from mainstream repositories (Brilmyer *et al.*, 2019), while Indigenous archival protocols for circulation complicate standard assumptions about public access to records (Christen and Pugh, 2011; Ghaddar, 2021; Poole, 2020). The prominence of these themes indicates a publication ecosystem in which social justice and the decentering of institutional authority have become well-established orientations in English-language archival scholarship.

Despite these divergences, both corpora converge on five major themes: digital archives and institutional management, policy dissemination and archival leadership, archival education and regional development, cultural memory and public platforms and organizational practices and project strategies. High-frequency keywords shared across both languages reveal a common emphasis on archival infrastructure and governance,

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Table 2. Topics and thematic keywords from Chinese- and English-language core archival journals

Topics	Chinese thematic keywords	English thematic keywords
Digital archives and institutional management	Archives, research, management, development, documents, electronic, archives institution, information, digital, application	Records, management, archives, archival, information, digital, research, archive, paper, study
Policy dissemination and archival leadership	Archives bureau, convene, ideology, promote, provincial party committee, director	Data, recordkeeping, systems, metadata, curation, system, electronic, government
Archival education and regional development	University, archival science, education, discipline, academia, second Sino-Japanese war (war of resistance against Japan)	Education, sector, implications, framework
Cultural memory and public platforms	Collection, platform, government, reflection, openness, art, implementation, people	Open, record, procurement, continuum, using, model, performance
Organizational practices and project strategy	Work, practice, enterprise, problems, project, strategy, utilization	Governance, iso, issues, practical, analysis
Cross-strait history and cultural identity	China, Taiwan, path, history, memory, culture, association, media	–
Community archives and preservation	–	Community media, preservation, social, theory, use
Indigenous rights and colonial critique	–	Indigenous, colonial, rights, human, justice, discourse, PSi archive, copyright, critical

Note(s): "–" indicates no direct counterpart topic in that corpus
Source(s): Authors' own elaboration

notwithstanding differences in ideological framing. Keywords such as "archives," "digital," "information," "management," "education," "system," "openness" and "strategy" reflect this shared focus. The recurrence of institutional terms such as "archives bureau," "government," "university" and "provincial party committee" across both corpora underscores the embeddedness of archival functions within formal administrative structures. From the perspective of information science and organizational studies, archives function as both instruments of documentation and repositories of institutional memory, playing a critical role in organizational continuity (Cook, 2013).

Keyword analysis

An analysis of keyword statistics from the collected publications identified the top ten keywords with the highest TF-IDF values, as presented in Table 3. However, keyword analysis of core Chinese archival journal articles presents certain limitations. Because Chinese journals do not consistently require authors to provide keywords, reliance solely on author-supplied keywords may be insufficient for thematic analysis. To address this limitation, the CKIP tagger, developed by Academia Sinica's Chinese Knowledge and Information Processing Lab, was used to segment abstracts and annotate POS tags. By extracting noun phrases from these abstracts, the 10 most frequent keywords were identified: archive(s), responsibility, archives administration, construction, society, nation, resource, culture and documents.

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Table 3. Top 10 keywords with the highest TF-IDF values in the Chinese- and English language corpora

No.	Chinese keywords	TF-IDF value	English keywords	TF-IDF value
1	Archive(s)	0.869	Records management	0.803
2	Responsibility	0.229	Archive(s)	0.269
3	Archives administration	0.138	Information management	0.231
4	Construction	0.122	Record	0.143
5	Information	0.117	Accountability	0.083
6	Society	0.085	Information governance	0.083
7	Nation	0.085	Information system	0.074
8	Resource	0.083	Management	0.074
9	Culture	0.082	Digital preservation	0.069
10	Document	0.079	Information	0.067

The thematic focus reflected in these keywords is shaped by the historical development of archival management and education systems in China, which were formalized in the 1980s. This development coincided with the emergence of the digital era in the 2000s and the enactment of the Archives Acts in Taiwan and China (Wu, 2019; Zhang and Yang, 2019). Consequently, keywords in Chinese archival literature emphasize the preservation of national histories and the role of national archives, with comparatively less attention to topics such as user information-seeking behavior or electronic archive construction.

For English-language core archival journals, the analysis included both author-supplied and database-generated keywords, such as keywords plus, system keywords and identifier keywords. The ten most frequent keywords in the English-language literature were records management, archive(s), information management, records, accountability, information governance, information systems, management, digital preservation and information. These keywords suggest that English-language archival research focuses on the management and governance of archives, records and information systems. Rather than emphasizing the foundational construction of archives, this literature tends to focus on refining and adapting existing systems to address challenges associated with digital transformation.

The study further identified five keyword clusters derived from core Chinese- and English-language archival journals, as presented in Table 4. Both coword networks indicated a shared emphasis on the development and management of electronic documents and records. However, differences remain in the developmental trajectories of journals published in the two languages.

Python was used to generate a social network graph of the top 50 keywords with the highest TF-IDF values. In this graph, clusters are organized according to thematic proximity, with edges between nodes indicating the strength of keyword co-occurrence. Each node represents a selected keyword. To maintain clarity, only the three strongest links for each node were retained in the visualization. The resulting keyword network visualizations are presented in Figures 4 and 5.

The following sections examine these keyword clusters in Chinese- and English-language core archival journals as distinct thematic focal points. Their co-occurrence patterns are analyzed to explore the relationship between the historical development of archival studies and evolving research orientations, viewed through the lens of DH.

In the Chinese-language corpus, Cluster 1 is organized around “information,” “resource(s)” and “construction,” pointing to a focus on building and operationalizing archival infrastructure through capacity-building and system-construction logics. In the

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Table 4. Keyword clusters for articles published in archival journals

	Chinese		English	
Clusters	Selected keywords	Avg. centrality	Selected keywords	Avg. centrality
Cluster 1	Information, resource(s) and construction	0.209 (high)	Management, records, archive (s), data, documentation,	0. 154 (high)
Cluster 2	Administration, responsibility, department, profession, problem	0.196 (medium)	Record, information management, archives management	0.083 (medium)
Cluster 3	Nation, theory, electronic document,	0.184 (medium)	System, governance, information technology	0. 067 (medium)
Cluster 4	Era, data, technique, resource, government	0.163 (medium)	Open data, data, retention, continuum	0.0571 (low)
Cluster 5	Society, nation, culture, China, socio-economy, education	0.158 (low)	Human rights, social justice, accountability, memory, trust, recordkeeping	0.068 (medium)

Source(s): Authors' own work

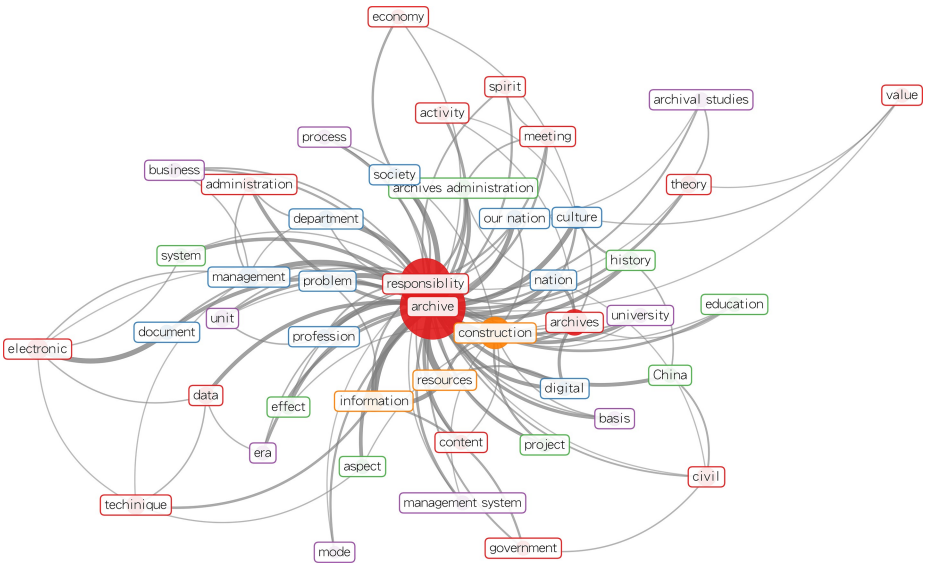


Figure 4. Keyword co-occurrence network for the Chinese-language corpus

Source: Authors' own work

English-language corpus, this cluster centers on "management," "records," "archives," "data" and "documentation," foregrounding the governance and management of existing records- and data-oriented systems.

Clusters 2 and 3 shift attention toward recordkeeping in e-government settings. In the Chinese-language corpus, the cluster structure foregrounds "archives administration," "nation" and "responsibility," framing e-government in terms of administrative mandates

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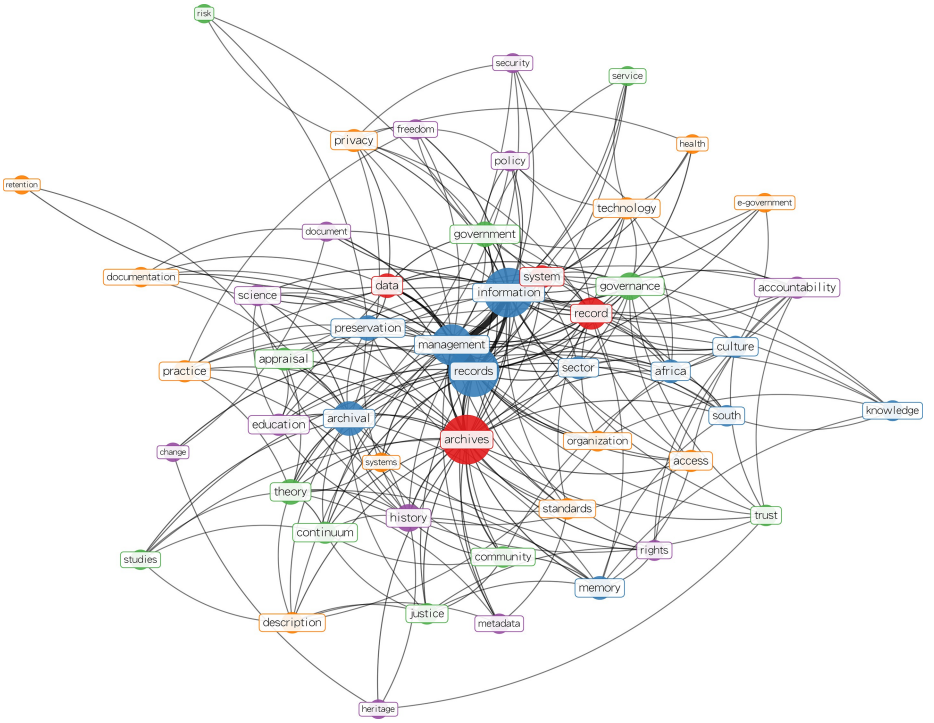


Figure 5. Keyword co-occurrence network for the English-language corpus
Source: Authors' own work

and institutional responsibilities (Zhang, 2006). In the English-language corpus, central nodes include "governance," "record" and "system," indicating recurring concerns about balancing information access with regulatory compliance. These principles have been operationalized in various governance contexts, including those relevant to e-governance systems in developing countries (Mnjama and Wamukoya, 2007). Taken together, these patterns suggest a structural divergence in how e-government is conceptualized across the two corpora: the Chinese-language corpus prioritizes the operational construction of digital archival systems, whereas the English-language corpus emphasizes the governance and accountability of existing digital records infrastructures.

Cluster 4 highlights an emerging area of divergence in how artificial intelligence (AI)-related developments are framed across the two corpora. In the Chinese-language corpus, this cluster foregrounds AI applications for declassification and document review, reflecting an operational orientation toward increasing processing efficiency and strengthening information security (Ma and Li, 2022). In the English-language corpus, Cluster 4 connects more loosely to open data and data retention, indicating a policy- and governance-oriented framing in which AI-enabled processing is less central as an organizing concept. This structural asymmetry suggests that AI–archival integration is an emerging research area in which the Chinese-language corpus currently exhibits a stronger operational focus than its English-language counterpart.

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Cluster 5 suggests the most pronounced divergence between the two language domains. In Chinese-language journals, keywords such as "society," "nation," "culture" and "document(s)" highlight the archival field's role in addressing social, national and cultural questions. However, the influence of this thematic cluster in Chinese journals appears to have declined over time.

Cluster 5 also exhibits the clearest cross-linguistic contrast and the most dynamic temporal pattern in the data set. In the Chinese-language corpus, "society," "nation" and "culture" remain closely linked to "archive(s)" across all four time periods Figure 6(a), indicating a stable macro-narrative orientation with little thematic drift over the study period. In contrast, the English-language corpus follows a markedly different trajectory Figure 6(b). Between 2014 and 2018, rights- and justice-oriented terms – such as "human rights" and "social justice", rose to prominence. In the most recent period (2019–2023), explicit references to "human rights" decline, while "accountability" rises to its highest observed level (TF-IDF ≈ 0.098). Taken together, these temporal patterns indicate a two-phase shift within the English-language corpus: an initial rights-and-justice orientation (2014–2018), followed by an accountability-oriented framing (2019–2023). By contrast, the Chinese-language pattern remains relatively stable across the full study period. This shift is reflected not only in the prominence of specific keywords but also in their distribution across time periods, indicating a measurable reconfiguration of thematic emphasis.

Discussion and conclusion

The following discussion treats the identified clusters as analytical proxies for thematic orientation. The interpretations presented here reflect publication patterns as mediated through computational analysis, rather than direct claims about underlying epistemological structures. All claims of cross-linguistic divergence are grounded in the cluster configurations documented above.

The findings collectively suggest that the Computational Turn (Berry, 2011) has manifested differently across Chinese- and English-language archival scholarship, producing not a unified methodological convergence but a structured divergence in how computational approaches are framed and prioritized. In the Chinese-language corpus, computational tools are predominantly

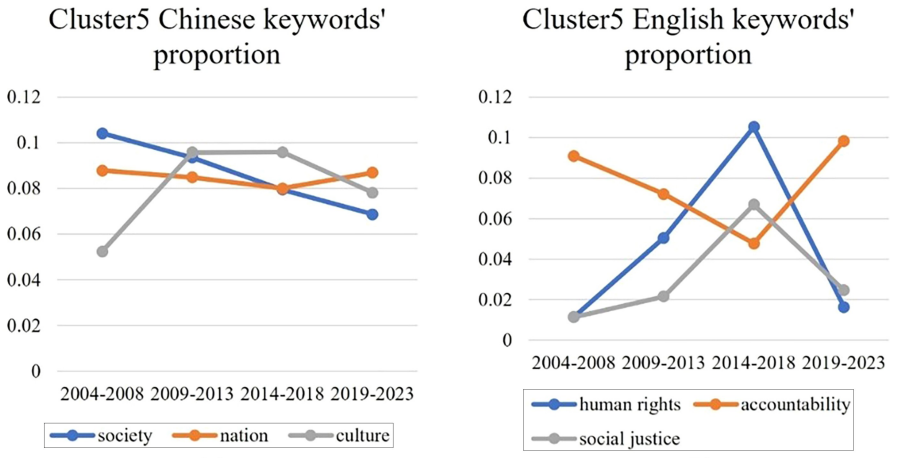


Figure 6. Cluster 5 Chinese and English keywords' proportion

Source: Authors' own work

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oriented toward operational capacity-building, including the construction of digital archival systems, AI-enabled processing and national infrastructure development. In the English-language corpus, the emphasis falls on the governance and accountability dimensions of existing digital records systems, with a discernible shift from a rights-and-justice framing toward an accountability-and-governance framing over the past decade.

These divergences indicate that the computational turn does not follow a single, globally convergent trajectory but is instead shaped by the institutional, cultural and political environments in which computational approaches are adopted. This finding extends Berry's (2011) framework by accounting for geopolitical and linguistic variation in how digital technologies reconfigure the objects and methods of archival knowledge production. Collectively, these observations challenge assumptions of uniform disciplinary convergence and position cross-linguistic bibliometric analysis as a productive method for identifying the structural conditions under which computational transformations emerge differently across scholarly communities.

At the same time, five shared thematic clusters indicate a common infrastructure-and-governance discourse that permeates both communities, despite their divergent conceptual orientations. The co-occurrence of shared keywords such as "archives," "digital," "management" and "education" across both corpora suggests that the institutional embeddedness of archives constitutes a durable point of convergence, even as epistemological priorities diverge (Cook, 2013).

The findings also highlight the ambivalence of platform-based digitization. On one hand, service-oriented, platform-based access can extend the reach and visibility of archival materials and foster broader participation in cultural and historical knowledge production. On the other hand, digitization and computational mediation introduce risks that are directly relevant to both scholarship and practice. A first risk is interpretive: large-scale automated analysis may privilege what is most visible, standardized or easily tokenized, potentially obscuring the contextual specificity that close, situated reading is designed to foreground. This concern is particularly evident in topic modeling, which has been critiqued for its tendency to flatten semantic complexity and obscure contextual nuance. A second risk is representational: when data sets are shaped by uneven collection histories, descriptive practices or access constraints, data-driven approaches may unintentionally amplify existing silences and further marginalize particular communities (Southerton, 2022).

These concerns suggest that the datafication of archives should be approached as a methodological and governance challenge rather than as a purely technical upgrade. As Ringel and Ribak (2024) argue, archival data are not "raw" but are constituted through infrastructures that carry interpretive and political consequences for what becomes visible or prioritized. Furthermore, while datafication can open new analytical and access pathways, it may also narrow the range of practicable research questions, particularly when computational priorities displace long-standing commitments to pluralism and social justice in archival work (Mordell, 2019).

Three implications follow from these findings for archives and information organizations navigating digital access and governance strategies. First, cross-linguistic divergence suggests that platform design and metadata policies should be evaluated not only for operational efficiency but also for representational accountability, recognizing that different scholarly communities may conceptualize archival authority and access in structurally different ways. Second, the prominence of open data, data retention and accountability-oriented terms in the English-language corpus highlights the need to align recordkeeping infrastructure with access governance and compliance requirements. Third, emerging trends toward AI-enabled review and declassification raise workforce and capacity considerations, indicating that technological adoption should be paired with robust oversight, auditability and transparency mechanisms.

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In conclusion, this study demonstrates that the computational turn in archival scholarship has not produced a uniform global trajectory but rather a structured divergence shaped by linguistic, institutional and cultural contexts. The Chinese-language corpus exhibits a sustained orientation toward operational digitization and national infrastructure, whereas the English-language corpus reflects an evolving engagement with governance, accountability and community-based archival ethics. Both trajectories underscore the field's capacity to mediate between computational efficiency and humanistic inquiry, highlighting the importance of maintaining this balance as a critical and reflexive practice rather than treating it as a purely technical given.

This methodological development has substantive implications for how archival studies positions itself at the intersection of technical infrastructure and interpretive inquiry. By enabling the comparative mapping of scholarly discourse across large corpora, computational approaches make visible forms of intellectual organization that close reading alone cannot capture. However, the interpretive, human-centered work of labeling, contextualizing and theorizing these patterns remains indispensable. The productive tension between distant reading and close interpretation – rather than being resolved in favor of either pole – is precisely what gives computationally assisted archival scholarship its analytical range (Neatrour *et al.*, 2018).

These divergent orientations suggest that what counts as the legitimate purpose of archival work and by extension, what computational approaches are mobilized to achieve, differs structurally across the two scholarly communities. The asymmetry between the two corpora is not merely a matter of language; rather, it reflects divergent conceptions of the purpose of archival scholarship, what constitutes methodological innovation and which problems are considered legitimate publishing priorities.

Although the present study reveals important findings, several limitations should be acknowledged. First, because the corpus was constructed using a keyword-based retrieval strategy, studies that engage computational approaches without explicitly using terms such as "DH" may not have been captured. This may result in the underrepresentation of methodologically innovative work that uses alternative terminology. Second, although the Chinese–English comparison provides a cross-linguistic analytical lens, it necessarily compresses substantial internal diversity within each language sphere, for instance, between Taiwan and Mainland China or among different Anglophone national traditions. Third, structural differences between Chinese and English present inherent preprocessing challenges: Chinese requires algorithmic word segmentation owing to its character-based writing system, whereas English relies on whitespace tokenization. These differences may introduce inconsistencies that may affect modeling outcomes. Although language-specific preprocessing pipelines and controlled vocabularies were applied to mitigate these effects, underlying linguistic divergence remains a constraint on direct comparison.

Future research could address these limitations by conducting region-specific analyses to test whether the patterns observed here hold across subregions and publication venues. It could also explore multilingual transformer-based approaches, such as Multilingual BERT (M-BERT) or Language-Agnostic SEntence Representations (LASER)-aligned word embeddings, with the aim of mitigating semantic asymmetries across language structures.

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Corresponding author
Chiao-Min Lin can be contacted at: cmlin@nccu.edu.tw

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